

JIEJIA LIU

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PERSONAL INFORMATION

E307, Physics Building
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Research interests:

Diffuse X-ray emission, Superbubbles, High-resolution spectroscopy, Instrumentation, Galaxy ecosystems.

EDUCATION

Ph.D. in Astronomy & Astrophysics

2021 – expected December 2026

Department of Astronomy, Tsinghua University

Advisor: Prof. Wei Cui

Bachelor of Physics

2017 – 2021

School of Physics, Nankai University

PUBLICATIONS

First Author

1. **Jiejia Liu**, Rui Huang, Chengzhe Li, et al., “X-rays from a newly discovered superbubble in M31”, 2024, [MNRAS](#), **533**, 3027.
2. **Jiejia Liu**, Sifan Wang, Hai Jin, et al., “Preliminary design of detector assembly for DIXE”, 2024, [JLTP](#), **216**, 273, [arXiv:2405.07559](#).
3. **Jiejia Liu**, Chunyang Jiang, Junjie Mao, et al., “Diffuse X-ray Explorer (DIXE): Sky survey strategy and collimator response demodulation”, 2026, [RAA](#), **26**, 125020, [arXiv:2606.10712](#).
4. **Jiejia Liu**, Junjie Mao, Shuinai Zhang, et al., “X-ray measurements of the elemental abundances for the diffuse emission in the nuclear starburst galaxy NGC 3079”, A&A submitted.

Co-Author

1. Hai Jin, Junjie Mao, . . . , **Jiejia Liu**, et al., “Diffuse X-ray Explorer: A high-resolution X-ray spectroscopic sky surveyor on the China Space Station”, 2024, [JLTP](#), **215**, 256, [arXiv:2406.09813](#).
2. Ruixuan Tian, Junjie Mao, **Jiejia Liu**, et al., “Simulation of non X-ray background for the Diffuse X-ray Explorer (DIXE) mission”, 2026, [Exp. Astron.](#), **61**, 17, [arXiv:2604.13569](#).
3. Yu Zhou, **Jiejia Liu**, Volodymyr Takhistov, Kazuhisa Mitsuda, “Probing ALP-Photon Mixing with High-Resolution X-ray Spectroscopy”, ApJ submitted, [arXiv:2507.22006](#).
4. Michael C. H. Yeung, Martin G. F. Mayer, . . . , **Jiejia Liu**, et al., “The SRG/eROSITA diffuse soft X-ray background II. spectra and morphology of the eROSITA bubbles in the western Galactic hemisphere”, 2026, A&A accepted, [arXiv:2605.02998](#).
5. Jiayi Chen, Haiwu Pan, . . . , **Jiejia Liu**, et al., “Modeling and in-flight characterization of the point spread function for the Einstein Probe Wide-field X-ray Telescope”, RAA submitted.

APPROVED PROPOSALS

Co-I, 300 ks on Einstein Probe Cycle 2 (ID: 0102)

2025

Title: The Largest Hot Superbubbles Ever Seen in a Low-Density Environment?

PI: Dr. Rui Huang

PRESENTATIONS

Talks

<i>X-rays from Superbubble Candidates in M31</i>	2023.12
The 2nd HUBS Workshop, Haikou, China	
<i>Exposure Map and Point-Spread Function of DIXE</i>	2024.12
The 2nd ShenKuo Symposium, Xiamen, China	
<i>X-rays from the superbubble in M31: a candidate for positive stellar feedbacks</i>	2025.03
Science of Diffuse Sources in the New Era, Nanjing, China	
<i>DIXE collimator response and algorithm implementation of DIXE imaging</i>	2025.06
Surveying the Hot Baryons Within the Milky Way ISSI-BJ Forum, Beijing, China	

Posters

<i>X-rays from Superbubble Candidates in M31</i>	2023.05
The 26th Guo Shoujing Conference, Huangshan, China	
<i>X-rays from Superbubble Candidates in M31</i>	2023.12
“Resolving Galaxy Ecosystem Across All Scales”, Hong Kong, China	
<i>Preliminary Design of Detector Assembly for DIXE</i>	2023.06
The 20th International Conference on Low Temperature Detectors, Daejeon, South Korea	
<i>Simulation of Magnetic Shielding for DIXE</i>	2024.06
SPIE Astronomical Telescopes + Instrumentation, Yokohama, Japan	

TEACHING EXPERIENCES

Teaching Assistant:

Major course, <i>Student Seminar</i>	2023
Major course, <i>Radiative Process in Astrophysics</i>	2024
Elective course, <i>Frontiers of Astrophysics</i>	2025

SCHOLARSHIPS & AWARDS

• Comprehensive Scholarship, First Prize, Tsinghua University	2025
• Comprehensive Scholarship, Second Prize, Tsinghua University	2022, 2024
• AMD Scholarship, Tsinghua University	2023
• “Gong Neng” Scholarship, Nankai University	2020
• Academic Excellence Scholarship, Nankai University	2018, 2019

REFERENCES

Prof. Wei Cui at Tsinghua University

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Prof. Junjie Mao at Tsinghua University

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Prof. Zhijie Qu at Tsinghua University

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Dr. Yu Zhou at National Astronomical Observatories

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